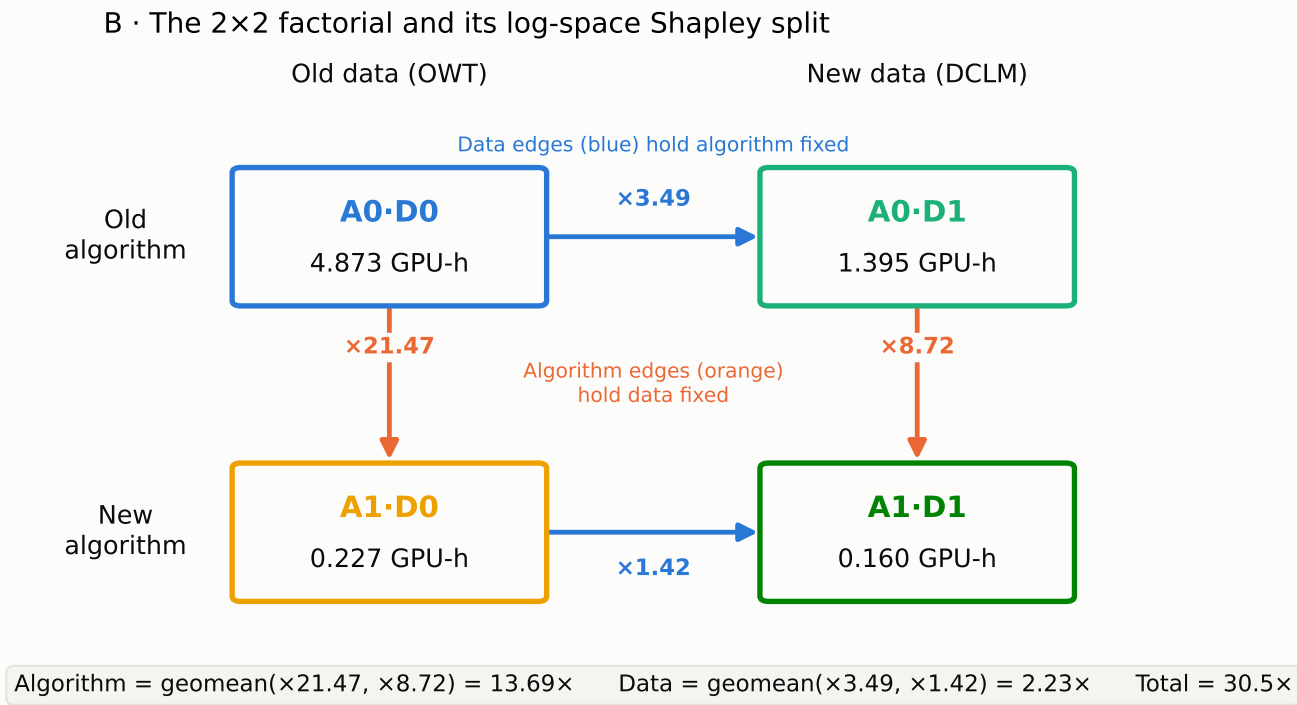
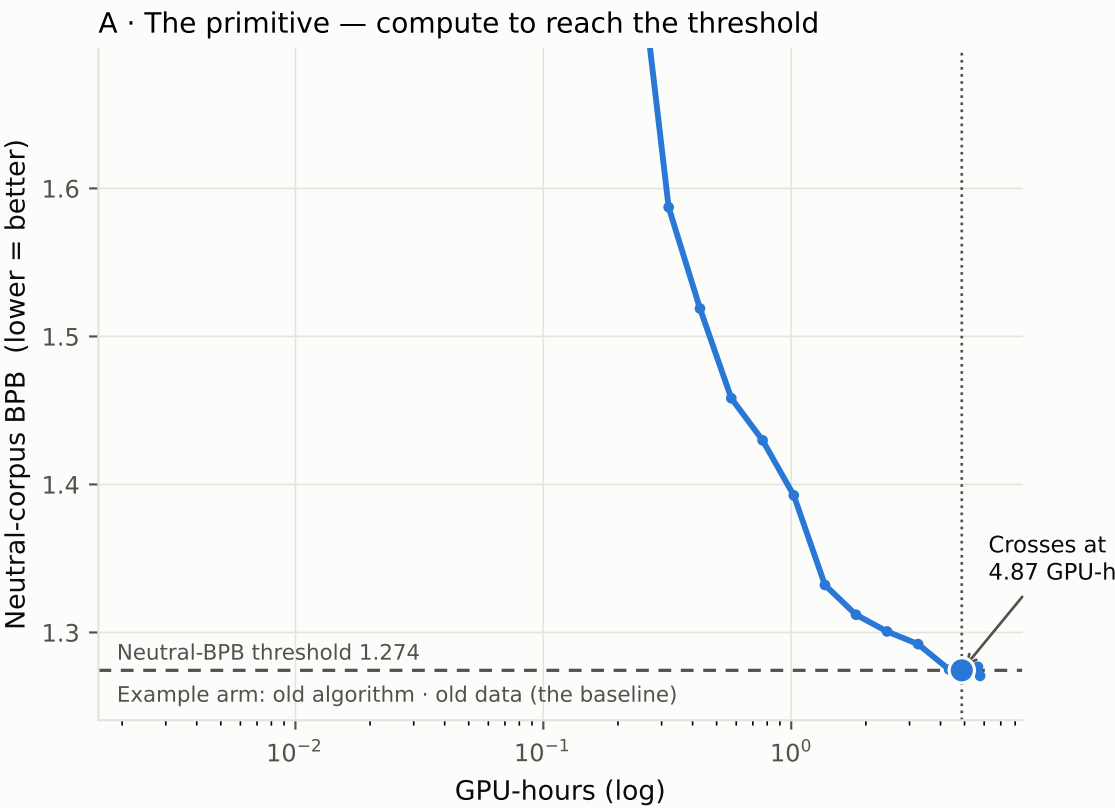


Factorial decomposition of compute-equivalent gain

Worked example: 124M GPT-2 baseline scale, current training recipe



Each edge is a GPU-hours ratio between two arms that differ in a single intervention; the Shapley multiplier for an intervention is the geometric mean of its two edges (both orderings averaged). Values re-derived from the four arm GPU-hours in results/small/ceg_newdef.json (held-out Wikipedia evaluation set, “wiki_eval”).